

[Skip to main content](#)

# MA 580-601 Spring 2026

## Numerical Analysis I Course Syllabus

[Click here for an HTML version of this syllabus](#)

### Table of Contents

- [Course Description](#)
- [Learning Objectives](#)
- [Course Structure](#)
- [Communication and Getting Help](#)
- [Textbook and other resources](#)
- [LaTeX resources](#)
- [Homework](#)
- [Test Information](#)
- [Classroom Expectations](#)
- [Grading](#)
- [Course Schedule](#)
- [Additional Information](#)
- [Digital Course Components](#)
- [Additional NC State Rules and Regulations](#)

### Course Information

- **Instructor:** Harshit Bhatt
- **Email:** [Email me at hbhatt@ncsu.edu](mailto:hbhatt@ncsu.edu)
- **Office:** 3149 SAS Hall
- **Office Hours:** Monday 3-4 pm (in person), Tuesday 1-2 pm (Zoom)
- **Course Modality:** This course is an online, asynchronous Distance Education course.
- **Course Website:** Find our course on [NC State Wolfware](#).

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Your day-to-day instructor for this course is Harshit Bhatt. Dr. Maulsby is the Coordinator for Distance Education (DE) and serves as the administrator for all DE courses in the Department of Mathematics, including this course.

The course videos were recorded by Dr. Tim Kelley *specifically* for this section of MA 580; he has no current involvement with the course. Any comments about homework, exams, grading, policies, etc., that are made in the videos applied to that semester, not to our semester.

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## Course Description

(3 credit hours) Algorithm behavior and applicability. Effect of roundoff errors, systems of linear equations and direct methods, least squares via Givens and Householder transformations, stationary and Krylov iterative methods, the conjugate gradient and GMRES methods, convergence of method.

Prerequisite MA 405; MA 425 or MA 511; high-level computer language

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## Learning Objectives

Students are expected to master techniques of Numerical Analysis. By the end of this course, students will be able to:

1. **Select appropriate numerical methods** for linear, nonlinear, and eigenvalue problems by analyzing problem structure, conditioning, stability, and computational constraints.
2. **Implement core numerical algorithms** for linear systems, least squares problems, nonlinear equations, eigenvalue problems, and multigrid methods, and evaluate their performance.
3. **Analyze sources of numerical error**, including floating-point effects, conditioning, and algorithmic instability, and use this analysis to diagnose method failure and guide method selection.
4. **Interpret convergence behavior and accuracy** of numerical methods using theoretical tools such as error bounds, residuals, and conditioning estimates.
5. **Use numerical software and computational tools responsibly**, understanding both their capabilities and limitations.
6. **Communicate numerical results professionally**, including clear documentation of assumptions, method choices, limitations, and observed behavior in computational experiments.

Homework and exams emphasize practical problem solving and computational reasoning, with theory used to justify algorithm choice, interpret results, and explain observed successes or failures.

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## Course Structure

### Lecture Materials

The material for this class is presented with 40 lectures divided up by week. Most weeks consist of three ~50 minute lectures. You may find it useful to follow a MWF calendar for this class, or you may choose to watch the lectures at other times each week.

## Live sessions

Will be decided later

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## Communication and Getting Help

### Forum

We will use **Piazza** for class discussion and homework questions. You are encouraged to use Piazza to discuss concepts and homework with discretion-hints and suggestions are okay, but full solutions will be deleted. You can post under your name or you may be anonymous to your classmates; however, the instructor will be able to see who you are. Please be respectful of your fellow classmates' ideas and attempts. If you have a math question you are not sure should be a public post, it is better to send me a direct message as a private post on Piazza.

Enroll in this class Piazza through the Piazza link near the top of Moodle.

Request: when asking a question about a specific item (like an item in Moodle), please include a direct link to that item (e.g. the Moodle url). It helps everyone answer your question a lot faster.

### Email

Contact me for confidential and private discussions about grades, scheduling office hours, etc. Please include MA 580 in the subject line. In general, I may not respond to messages outside of business hours (M-F, 9am-5pm), and it may take 24-72 hours for me to respond to an message. Math questions are generally best asked in office hours or on our forum, not by email.

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## Textbook and other resources

We will use free online textbooks which are provided on Moodle and which can also be found through [NC State Library Course Reserves](#).

- I. C. F. Ipsen, Numerical Matrix Analysis, SIAM, 2009. [Link](#)
- C. T. Kelley, Iterative Methods for Linear and Nonlinear Equations, SIAM 1995. [Link](#)
- Desmond J. Higham and Nicholas J Higham, Matlab Guide: Second Edition, 2005. [Link](#)

## LaTeX Resources

### LaTeX resources

The links below provide learning materials and software for LaTeX. Additional resources are located in Moodle.

- [Learning LaTeX — Griffiths & Higham \(NCSU library\)](#) — Library record for the book "Learning LaTeX".
  - [LaTeX Project — software and documentation](#) — Official LaTeX distribution and documentation.
  - [LaTeX Wikibook — comprehensive guide](#) — Community-maintained LaTeX reference and examples.
  - [Getting started with LaTeX — primer \(Trinity College Dublin\)](#) — Introductory tutorial and examples.
  - [MiKTeX — easy-to-install LaTeX for Windows](#) — Windows installer and documentation for MiKTeX.
  - [Using TeX notation in Moodle forums](#) — Moodle documentation for inserting math with TeX.
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## Homework

This course will have five written homework assignments. You will be required to use LaTeX or another program to type up your answers. Keep in mind that writing solutions can be challenging and require a lot of time. You should be sure to begin the next homework set as soon as possible.

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## Test Information

### Test Dates

- Test 1: 23 - 24 Feb
- Test 2: 09 - 10 April
- Final Exam: 30 April - 01 May

### Proctors

All examinations will be proctored through DELTA. You will either take your exam with DELTA, or coordinate remote proctoring with DELTA, according to which situation best describes you:

- If you live less than 50 miles away from Raleigh, NC, then you will take your tests on our campus through the Distance Education Testing Centers. Students should be

# NC STATE UNIVERSITY

mindful of closing hours for both Testing Centers, and give themselves plenty of time to complete their exams. For information, please visit [DELTA on-campus testing services](#).

- Those students who live more than 50 miles away from Raleigh, NC do not have to take their tests on NCSU campus. They may use a proctor in their town for testing. The proctor must be approved in advance through DELTA (not the instructor). It can take up to 1 week to verify a proctor and set up all needed contact info, so please do this early! Please visit the remote proctor website at [DELTA remote proctoring](#).

I encourage you to look through DELTA Testing Service's website, especially their [frequently asked questions \(FAQs\)](#). Please note that if you are taking a paper exam at DELTA, it will be administered on an iPad.

**Note:** the word *remote* in this context means that you are using a professional testing site (for example, a testing center, a local library, a college, etc.) other than DELTA. Remote testing does not mean testing at home.

## Quick Links

Please sign up your preferred time and date as soon as possible. Time slots fill up fast:

## DELTA and exam links

- [Your Final Exam Schedule](#)
- [On-Campus Testing \(DELTA\)](#)
- [Off-Campus Proctoring](#)
- [Reschedule with DELTA](#)

You may reschedule an exam appointment directly with DELTA using the link above. You do not need to contact the instructor for permission, provided that your new appointment is scheduled within the exam's posted date range.

## Calculators

You may use basic (four-function) calculators on exams. These calculators must have no calculus or graphing capabilities.

There are descriptions and examples of calculators here: [Calculator Guidelines](#). If you are on-campus, you may rent a suitable calculator from the NC State Libraries: [Calculator rentals at NC State Libraries](#).

## Test Format

Each exam will be a written, in-person exam administered at an approved testing location. You must bring appropriate photo identification and any permitted materials specified in this syllabus and in my course announcements.

If you take the written exam at DELTA, then you will complete it on an iPad with an Apple Pencil.

## Regrading

If a grading error is found after looking at the posted answer key then you should provide a written explanation of the error attached to the original test or submit a regrade request through Gradescope, to the instructor within 1 week. Do not alter the original work. The entire test may be re-graded and the test grade is subject to remain the same, increase or decrease at the discretion of the instructor.

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## Classroom Expectations

1. **Course Structure:** This is an online class; you are responsible for your own learning and for pacing yourself within course guidelines.
  - Watch the video lectures scheduled each week and follow the Course Calendar.
  - Track all due dates in Moodle and on the Course Calendar.
  - Complete all assignments in a timely fashion.
  - Post your math questions in the course forum for discussions and Q&A.
  - Optional: attend live sessions.
2. **Communication and announcements:** Please check your email, the course forum, and the Moodle site regularly. All announcements sent by email will also be saved under Announcements on Moodle. You are responsible for knowing the content of course emails.
3. **Respect and professionalism:** Treat everyone in class (students and instructor) with respect and courtesy. Be active and prepared in any live sessions. Come to office hours ready to ask questions and communicate with others.
4. **Accountability:** You are responsible for resolving any confusion about assignments, due dates, exams, accommodations, etc., in a prompt manner.
5. **Academic integrity:** Do not submit work that is not yours. It is understood that your name on any assignment indicates your adherence to the NC State Honor Pledge: "I have neither given nor received unauthorized aid on this test or assignment."
6. **Exam device policy:** Review the permitted items before each exam. Keep phones and other forbidden devices powered off and stored away during exams to avoid accidental use.

## Student Success

Student well-being is important to success at NC State. Every student, faculty member, and staff member enriches the community through varied perspectives, knowledge, and experience. Our classroom should be a space where every student is respected and heard.

In an effort to affirm and respect the identities of all students in the classroom and beyond, please contact me if you wish to be referred to using a name and/or pronouns other than those listed in the student directory.

I welcome any suggestions you have for making our classroom more welcoming.

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## Grading

### Grade Weighting and Numerical Conversion

Your grade will be determined by the following breakdown:

- Homework: 40%
- Tests: 30% (15% each)
- Final: 30%

Grades are tracked in real-time in the Moodle **Gradebook**.

### Conversion from Numerical Grade to Letter Grade

A student's numerical average will be converted to a letter grade as follows (do not expect any additional rounding, extra credit, or curves):

Standard Conversion Table

| Grade | Range    |
|-------|----------|
| A+    | 97-100   |
| A     | 93-96.99 |
| A-    | 90-92.99 |
| B+    | 87-89.99 |
| B     | 83-86.99 |
| B-    | 80-82.99 |
| C+    | 77-79.99 |
| C     | 73-76.99 |
| C-    | 70-72.99 |
| D+    | 67-69.99 |
| D     | 63-66.99 |
| D-    | 60-62.99 |
| F     | 0-59     |

## Requirements for Credit-Only (S/U) Grading

In order to receive a grade of S, students are required to take all exams, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to [REG 02.20.15 - Credit-Only Courses](#).

It is the student's responsibility to check if an S grade gives progress towards their degree(s).

## Requirements for Auditors (AU)

Information about and requirements for auditing a course can be found at [REG 02.20.04 - Audits](#).

## Policies on Incomplete Grades

### NC State Policy

At the discretion of the instructor, students may be given an IN grade for work not completed because of a serious interruption in their work not caused by their own negligence. An IN must not be used, however, as a substitute for an F when the student's performance in the course is deserving of failing. An IN is only appropriate when the student's record in the course is such that the successful completion of particular assignments, projects, or tests missed as a result of a documented serious event would enable that student to pass the course.

If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student. See the university policy on incomplete grades: [REG 02.50.03 - Grades and Grade Point Average](#).

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## Course Schedule

The course schedule is tentative and subject to change. Adjustments may be made to accommodate the pace of the class and unforeseen circumstances. All major changes will be announced in class and posted on Moodle.

- Course Calendar: [HTML](#)



## Important Dates

For holidays and other university closures, please consult [the general NC State Academic Calendar](#).

Your final exam schedule is already determined; find it here: [NC State Final Exam Calendar](#). You are responsible for reviewing your final exam sessions to arrange a suitable time for this course.

## List of Topics

Here are the major topics of MA 580, with the approximate regular-semester time allocated to each:

- **Preliminaries, Floating Point Arithmetic, and Conditioning (~1.5 weeks).** Major topics include review of mathematical prerequisites, notation and preliminaries, floating-point arithmetic, roundoff error, conditioning and stability, and an initial introduction to linear systems and Gaussian elimination.
- **Direct Methods for Linear Systems (~2 weeks).** Major topics include Gaussian elimination in depth, the factor–solve paradigm, LU and PLU factorizations, pivoting strategies, error analysis in direct solvers, the Banach lemma, and special matrix structures including symmetric positive definite systems and Cholesky decomposition.
- **Sparse Linear Systems and Model Problems (~1.5 weeks).** Major topics include sparse matrix structure, Poisson’s equation, sparse Gaussian elimination, time-dependent problems, Crank–Nicolson methods, and applications such as the time-dependent Schrödinger equation.
- **Least Squares Problems and Orthogonal Factorizations (~2 weeks).** Major topics include linear least squares problems, QR decomposition, Householder transformations, Givens rotations, classical and modified Gram–Schmidt methods, and the singular value decomposition (SVD), including rank-deficient and underdetermined systems.
- **Stationary Iterative Methods (~1 week).** Major topics include stationary iterative methods for linear systems, classical schemes such as Jacobi and Gauss–Seidel, convergence considerations, and applied examples including the linear neutron transport equation.
- **Krylov Subspace Methods (~2 weeks).** Major topics include Krylov subspace methods, GMRES and its analysis, implementation and preconditioning strategies, conjugate gradient methods (CG, CGNR, CGNE), and applications to large-scale linear systems.
- **Nonlinear Equations and Newton-Type Methods (~2 weeks).** Major topics include nonlinear systems of equations, contraction mappings, Newton’s method and convergence theory, Newton–Krylov methods, implementation issues, and practical nonlinear solver examples.
- **Nonlinear Least Squares and Eigenvalue Problems (~1.5 weeks).** Major topics include nonlinear least squares problems, Gauss–Newton iteration, eigenvalue

problems, the power and inverse power methods, shifted iterations, eigenvalue conditioning, and the Bauer–Fike theorem.

- **Error Control and Multigrid Methods (~2 weeks).** Major topics include residual correction and iterative refinement, two-grid methods, multigrid V-cycles, multigrid as a preconditioner, nonlinear multigrid ideas, and multigrid methods for integral equations.

## Course Continuity

To ensure course continuity, changes made to the method of instructional delivery, course structure, course schedule, number of assignments, grading or other aspects of the course after the start of the term will be communicated to all students in written form (e.g., by an instructor announcement) when course changes are implemented.

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## Additional Information

### Student Expenses

### Late Assignments

Generally no late assignments are accepted in this course. Please plan your work carefully and submit assignments by their stated deadlines. Extensions may be occasionally granted by the instructor for extenuating circumstances, which are best documented with [NC State Absence Verification](#).

For longer-term, recurring, or more serious illness or other interruptions to your participation in this class, you should reach out to your instructor as soon as you can.

### Late Examinations

*Excused absence.* If an exam is missed with an excused absence (that is, for a university-approved reason with supporting documentation), then a make-up test will be scheduled individually. The make-up test may contain different questions and be assessed differently than the regular test. Documentation for an excused absence must be provided within 1 week of the missed class. All absences that require a make-up exam or other special accommodations must go through the NC State University absence verification process. Here is the link to that office: [NC State Absence Verification](#).

*Failure to schedule.* You must schedule exams in a timely fashion to guarantee that you will be able to take them. It is the instructor's discretion whether a make-up exam will be allowed if you are not able to schedule an exam appointment within the selected time frame. If the instructor approves a make-up exam, there may be a 10% penalty on that exam. The make-up test may contain different questions and be assessed differently than the regular test.

*Other absences.* If an exam is missed for an unexcused absence, that exam will be given a score of 0.

## Attendance

Since this course is an asynchronous online course, there is no daily attendance. Instead, each student's participation in Moodle is tracked to check for regular activity. For complete attendance and excused absence policies, please see Attendance Regulations ([NCSU REG 02.20.03](#)).

## Academic Integrity

Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct ([NCSU POL11.35.01](#)). Your submission of any exam indicates "I have neither given nor received unauthorized aid on this test or assignment." Violations of academic integrity will be handled in accordance with the Student Discipline Procedures ([NCSU REG 11.35.02](#)).

Posting any course material to websites like Chegg, ChatGPT, and Course Hero is a violation of copyright law and course policy and is strictly prohibited. Violations of this policy will be reported to the [Office of Student Conduct](#).

- Tests: Proctored tests are closed book assessments. You may not consult any internet resources nor receive help from anyone else. Do not share information about the content on the exams with anyone else in the class. See the calculator policy in the Test Information section of this syllabus.
- Homework: You may consult your notes, the textbook, each other, or online resources.
- Forum discussion boards: You are encouraged to discuss mathematical concepts and problems with your classmates. However, you must arrive at your own solutions with your own work. Do not seek nor state final answers on the forum; focus on understanding the concepts.

## Disability Resources

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Resource Office at Holmes Hall, Suite 304, 2751 Cates Avenue, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation ([NCSU REG 02.20.01](#)).

Students who receive accommodations are responsible for filing those accommodations officially with DRO before the tests occur. It is not appropriate to tell the instructor that you should receive accommodations without going through the official channels. The instructor

will only adjust test conditions after receiving the official Accommodation Letter Notification from DRO. Adjustments are never made retroactively to past examinations.

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## Digital Course Components

Because of the way our course is structured, students need internet connectivity in order to read course materials and complete assignments. NC State's Office of Information Technology (OIT) provides [technology guidance](#), including requirements and recommendations for computer hardware.

For access to computing hardware, please see the NC State University Libraries [Technology Lending](#) and the general [Library Computing resources](#). There are several computers available for use around campus, including in the [Mathematics Tutoring Center](#).

Digitally hosted course components will include but are not limited to Moodle and Zoom.

## Software

There are several resources available to assist students with technical or computer issues. For technology help, please consult the [Office of Information Technology](#) at NC State University.

Here are some of the primary applications commonly used in online mathematics courses:

- [Moodle and Wolfware](#): Our course is hosted online on Moodle, where you can find the course content, a link to this syllabus, and the gradebook.
  - [Moodle Accessibility Statement](#)
  - [Moodle Privacy Notice](#)
  - [NC State WolfWare Privacy Statement](#)
- [Panopto](#): NC State uses Panopto for video hosting.
  - [Panopto Accessibility Features](#)
  - [Panopto Privacy Policy](#)
  - [Panopto Support](#)
- [Piazza](#): this site hosts our class forum. You will use **Piazza** to ask questions regarding the lecture or homework.
  - [Piazza Accessibility Statement](#)
  - [Piazza Privacy Policy](#)
  - [Piazza Support Center](#)
- [Google Meet](#): when needed.
  - [Google Meet Accessibility features](#)
  - [Google Meet Security and Privacy](#)
  - [Google Meet Help](#)
- [Zoom](#): when needed.
  - [Zoom Accessibility Statement](#)

- [Zoom Privacy Policy](#)
- [Zoom Support](#)
- [Gradescope](#): We will use Gradescope for collecting and returning written work.
  - [Gradescope Accessibility Information](#)
  - [Gradescope Privacy Policy](#)
  - [Gradescope Help Center](#)

You must address the accessibility of these websites for yourself during the course drop/add period.

The instructor is not responsible for ensuring privacy or accessibility of electronic materials that are not required components of the course (e.g., links to supplemental information that is not part of the required reading list). However, the instructor will judiciously consider the privacy, copyright, and accessibility of supplemental links provided to students and warn them of any known issues or concerns in this regard. See Online Course Material Host Requirements ([NCSU REG 08.00.11](#)).

## Electronically Hosted Components

Please be advised that live meetings for this course may be recorded for current and potential future educational purposes. By your continued participation in this recorded course, you are providing your permission to be recorded. If you would like for your likeness to be edited out of a recorded video, please contact me and I will edit the video accordingly.

## Required Statement

Students may be required to disclose personally identifiable information to other students in the course, via digital tools, such as email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

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## Additional NC State Rules and Regulations

### Your rights and responsibilities

Students are responsible for reviewing the NC State University Policies, Rules, and Regulations (PRRs), which pertain to their course rights and responsibilities, including, but not limited to:

- [POL 04.25.05](#) Opportunity and Non-Discrimination Policy Statement
  - [REG 02.20.01](#) Academic Accommodations for Students with Disabilities
- [POL 11.35.01](#) Student Conduct
  - [REG 11.35.05](#) Code of Student Conduct

# NC STATE UNIVERSITY

- [REG 11.35.02](#) Student Discipline Procedures
- [REG 02.20.03](#) Attendance Regulations
- [REG 02.50.03](#) Grades and Grade Point Average
- [REG 02.20.15](#) Credit-Only Courses
- [REG 02.20.04](#) Audits
- [REG 08.00.11](#) Online Course Material Host Requirements

You may also find additional references at [NC State Office of Equal Opportunity](#).

## Non-Discrimination Policy

NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at

[POL 04.25.05 - Equal Opportunity and Nondiscrimination Policy](#) or [the Office of Equal Opportunity](#)

Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

## Support

Everyone is encouraged to take care of themselves and their peers. If you need additional support, there are many resources on campus to help you:

1. [Counseling Center](#)
2. [Health Center](#)
3. [Share a Concern](#)
4. [Pack Essentials](#)

## Course Evaluations

ClassEval is the end-of-semester survey for students to evaluate instruction of all university classes. The current survey is administered online and includes 12 closed-ended questions

and 3 open-ended questions. Deans, department heads, and instructors may add a limited number of their own questions to these 15 common-core questions.

Each semester students' responses are compiled into a ClassEval report for every instructor and class. Instructors use the evaluations to improve instruction and include them in their promotion and tenure dossiers, while department heads use them in annual reviews. The reports are included in instructors' personnel files and are considered confidential.

Online class evaluations will be available for students to complete during the last two weeks of the semester for full semester courses and the last week of shorter sessions. Students will receive an email directing them to a website to complete class evaluations. These become unavailable at 8am on the first day of finals.

- [Contact ClassEval Help Desk](#)
- [ClassEval website](#)
- [Information about ClassEval and how the information is used](#)

## Required statement

This course engages diverse scholarly perspectives to develop critical thinking, analysis, and debate and inclusion of a reading does not imply endorsement. *This statement is required per [UNC Policy Manual 400.1.6](#), adopted 12/19/2025.*

## Syllabus Modification Statement

Our syllabus represents a flexible agreement. It outlines the topics we will cover and the order we will cover them in. Minor changes in the syllabus can occur if we need to slow down or speed up the pace of instruction.

Per [REG 02.20.07 – Course Syllabus](#), syllabi do **not** constitute an express or implied contract among the student, faculty, or institution. Rather, syllabi serve as a guide for the course.

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This syllabus was designed by Bevin Maultsby to meet the NC State University standards [REG 02.20.07 - Course Syllabus](#) (according to the February 3, 2026 revision) and to be WCAG 2.1 AA compliant.